Quadratic Equations

A Quadratic Equation has max 2-degree term.

General Quadratic Equation

$$\underbrace{ax^2}_{\text{quadratic term}} + \underbrace{bx}_{\text{linear term}} + \underbrace{c}_{\text{constant term}} = 0$$

* a, b, c are called coefficients.

Quadratic Equations

Quadratic Equations can be solved in many ways!

1 Factorizing

- $a = 1$
- $a \neq 1$

2 Complete The Square

- $a = 1$
- $a \neq 1$

3 The Quadratic Formula

1 FACTORIZING (a = 1)

Some quadratic equations can be factorized.

$$x^2 + bx + c = 0$$

- bx: $-$ sum of solutions
- c: product of solutions

Find 2 numbers that sum to b and their product is c.

1 FACTORIZING (a = 1)

Solve $x^2 - 2x - 24 = 0$

Sol^{n}//

$$x^2 - 2x - 24$$

$$= (x + 4)(x - 6) = 0$$

$$x = -4 \quad \text{OR//} \quad x = 6$$

factor pairs of -24	sum
-1, 24	23
1, 24	-23
-2, 12	10
2, -12	-10
3, -8	5
-3, 8	-5
4, -6	-2
-4, 6	2

b!

* You won't need the table when you've practiced!

1 FACTORIZING ($a \neq 1$)

Factorizing can be tougher when $a \neq 1$!

$$ax^2 + bx + c = 0$$

We give you two methods to help.

1 Factorizing ($a \neq 1$) Method 1

1. Find Factor Pairs Of $a \times c$
2. Sum Factor Pairs Till You Get b
3. Split b Into Factor Pairs
4. Collect Like Terms

1 FACTORIZING $(a \neq 1)$ METHOD 1

Solve $6x^2 - 7x - 5 = 0$.

Sol^{n} 1 $6 \times -5 = -30$

2

-30	sum
$1, -30$	-29
$-1, 30$	29
$2, -15$	-13
$-2, 15$	13
$3, -10$	-7 → b
$-3, 10$	7
$5, -6$	-1
$-5, 6$	1

3 $6x^2 - 7x - 5$

$6x^2 + 3x - 10x - 5$

4 $3x(2x+1) - 5(2x+1)$

$(3x - 5)(2x + 1) = 0$

$x = 5/3$ OR $x = -1/2$.

1 FACTORIZING $(a \neq 1)$ METHOD 2

1. SOLVE $x^2 + bx + ac = 0$
2. FACTOR
3. DIVIDE BOTH SOLUTIONS BY a

1 FACTORIZING ($a \neq 1$) METHOD 2

Solve $6x^2 - 7x - 5 = 0$.

Sol^{n}//

1 $x^2 - 7x - 30 = 0$

2 $(x + 3)(x - 10) = 0$

3 $x = -\frac{3}{6}$ OR $x = \frac{10}{6}$

$= -1/2$ $= 5/3$

* You can get the factors back from the solutions!

$x + \frac{1}{2} \rightarrow 2x + 1$ $x - 5/3 \rightarrow 3x + 5$

2 Complete The Square $(a = 1)$

Sometimes, we can't factorize.

However, we can always turn our quadratic into a

$$a(x + p) + q = 0$$

Completing The Square

2 Complete The Square $(a = 1)$

We complete the square by :

1. Finding $\frac{b}{2}$
2. Subtracting & Adding $\left(\frac{b}{2}\right)^2$
3. Factor & Simplify

2 Complete The Square $(a = 1)$

Solve $x^2 - 2x - 24 = 0$

Sol

1 $\frac{b}{2} = -1$, $\left(\frac{b}{2}\right)^2 = 1$

2 $x^2 - 2x + 1 - 1 - 24$

3 $(x-1)^2 - 25 = 0$

$(x-1)^2 = 25$

$x - 1 = \pm 5$

$x = 1 \pm 5 = 6$ OR -4

2 Complete The Square ($a \neq 1$)

For $a \neq 1$, completing the square is the same. But, we factor a first.

2 Complete The Square $(a \neq 1)$

Complete the square $3x^2 + 2x - 2$

Sol^{n}//

$$3\left(x^2 + \frac{2}{3}x\right) - 2$$

$$3\left(x^2 + \frac{2}{3}x + \frac{1}{9} - \frac{1}{9}\right) - 2$$

$$3\left(x + \frac{1}{3}\right)^2 - \frac{1}{3} - 2$$

$$3\left(x + 1/3\right)^2 - 2\,1/3$$

* You can solve by algebra for solutions.

3 The Quadratic Formula

The quadratic formula solves any quadratic equation!

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

3 The Quadratic Formula

Solve $6x^2 - 7x - 5 = 0$.

Sol^{n}

$$x = \frac{-(-7) \pm \sqrt{(-7)^2 - 4(6)(-5)}}{2(6)}$$

$$= \frac{7 \pm \sqrt{49 + 120}}{12}$$

$$= \frac{7 \pm 13}{12}$$

$$x = \frac{7+13}{12} = \frac{20}{12} = \frac{5}{3} \quad \text{OR} \quad x = \frac{7-13}{12} = \frac{-6}{12} = -\frac{1}{2}$$